

Ganesh Iyer



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Education

Sep 2025–2027 **ETH Zürich**, *Master of Science*,
Robotics, Systems and Control
GPA **5.72/6**
Nov 2021–May 2025 **Indian Institute of Technology Bombay**, *Bachelor of Technology*,
Mechanical Engineering (with Honors) specializing in Systems and Controls,
Minor in Artificial Intelligence and Data Science
GPA **9.87/10**

Publications

Conference Paper Lin Yang, **Ganesh Iyer**, Baichuan Lou, Sri Harsha Turlapati, Chen Lv, Domenico Campolo, "**Path Planning in Complex Environments with Superquadrics and Voronoi-Based Orientation**", IEEE Conference on Robotics and Biomimetics 2025

Skills

Languages Python, MATLAB, C++
Software ROS2, NVIDIA Isaac Lab/Sim/Gym, Git, GitHub, Docker, L^AT_EX
Libraries OpenCV, Pandas, Pytorch, NumPy, Scikit Learn

Internships

June 2025 – **Reinforcement Learning for Quadruped Robot Locomotion**
July 2025 [xTerra Robotics, India](#)

- Researched **quadruped locomotion control**, combining model- and learning-based methods
- Trained RL policy for Svan M2 via **guided policy search** with trajectory tracking rewards, using data from **trajectory planner** generated offline, resulting in accelerated training
- Used PPO training in NVIDIA Isaac Gym with domain randomization for Sim2Real, validated in **MuJoCo**
- Trained NN-based **state estimator** for base linear, angular velocity and base height estimation from sensor data buffer, required by the RL Policy
- Deployed on Svan M2 **hardware**, achieving **trot gait** (0.7 m/s forward, 0.5 m/s lateral, 1 rad/s yaw)
- Video link of training, simulation and hardware results can be found [here](#)

May 2024 – **Voronoi Diagram Based Path Planning for Superquadric Obstacles**
Jul 2024 Global Connect Fellowship, Nanyang Technological University | Guide: [Prof Domenico Campolo](#)

- Conducted a comprehensive survey on **superquadrics** and their role in obstacle-based **path planning**.
- Developed a novel algorithm to generate **Voronoi diagrams** for superquadrics and applied **Dijkstra's** algorithm on the resulting graph for efficient path planning.
- Used **image processing** to extract object **point cloud** data from an environment image, fitted superquadrics onto obstacles, and tested the algorithm on an experimental setup, using objects of various shapes as maze
- Benchmarked the Voronoi-based planner against **classical methods** like A* and RRT*, and **heuristic** planners such as genetic algorithms, and a recent state-of-the-art drone planning algorithm in 3D.
- Concluded that Voronoi planning for superquadrics offers **safer**, shorter paths with fewer nodes, **low computational space**, requiring only object **point cloud** from image, and no other parameters.

Scholastic Achievements

2024 **Department Rank 2** out of **196** students in Mechanical Engineering, IITB
2023 Awarded the **Institute Academic Prize two times**, for exceptional academic performance in the second and third year among Mechanical Engineering students
Recipient of the **Undergraduate Research Award** for exemplary research work in the second year summer

- Achieved perfect **10** Semester Performance Index in **five semesters**: Semester **3,4,5,7** and **8**
 Awarded the **Academic Proficiency(AP)** grade for **exceptional academic performance** in **7** courses
- 2021 Secured **All India Rank 179** in **JEE Main** among 1.2 million aspirants
 Secured **All India Rank 1234** in **JEE Advanced** among 250,000 aspirants

Projects

- Feb 2026 – **Humanoid Motion Control for Soccer**
 Present NomadZ – RoboCup Team, ETH Zürich | nomadz.ethz.ch
- Developing **imitation learning** based motion control for **Booster K1** humanoid robots for RoboCup soccer
 - Retargeted custom-recorded human motion videos (**walking, dribbling, kicking**) onto the robot using **GVHMR** and **GMR** for human-to-robot **motion retargeting**
 - Trained a **velocity-tracking locomotion policy** and a **dribbling policy** leveraging **Adversarial Motion Priors (AMP)** to encode natural style while satisfying the task objective, using **NVIDIA Isaac Lab** (MimicKit)
 - Building **deployment** code for **Sim2Real** transfer and integrating the learned policies with the team's **vision** and **behaviour** pipelines
- Dec 2025 – **Monocular Visual Odometry**
 Jan 2026 Mini Project | [VAMR 2025](https://vamar2025.github.io)
- Built a monocular visual odometry pipeline (Python/OpenCV) using **Harris** corners + **SIFT**, **KLT optical flow** tracking, **PnP** + **RANSAC** pose estimation, triangulation to recover 6-DoF camera poses
 - Added robust bootstrapping, nonlinear **pose refinement** (Levenberg–Marquardt), re-initialization, and landmark rescaling to handle drift, feature loss, and scale ambiguity
 - Tuned and validated across KITTI, Malaga, Parking, and a custom ETH Polybahn ride dataset, achieving **locally consistent** trajectories. Results can be viewed [here](#).
- Aug 2024 – **Koopman/DMD based Control for Non-Linear Dynamical Systems**
 Apr 2025 Bachelor Thesis Project, IIT Bombay | Guide: [Prof. Abhishek Gupta](#)
- Extensively Studied **Koopman theory**, and many data-driven control strategies, including Dynamic Mode Decomposition(**DMD** and **EDMD**), and its control variants(**DMDc**), and **Hankel-based MPC**
 - Applied Koopman Theory on simple problems, such as spring- mass systems, vanderpol oscillators, pendulum for predicting dynamics, and applying to more complex problems such as a **two-link manipulator**
 - Applied Koopman/DMD techniques on the simulation data for a planar soft robot for getting **approximate linear dynamics** in the **lifted space**
- Aug 2024 – **Warehouse Drone - E-Yantra, Nation-wide Robotics competition**
 Sep 2024 Competiton | [eYRC portal](https://eYRC.org)
- Developed PID control algorithm for roll, pitch,yaw and throttle control of a Swift Pico Drone in Gazebo, with the node files and communication pipeline in ROS2 environment
 - Used **A*** as the **motion planning** algorithm to **navigate** and simultaneously control the drone within a **warehouse** environment in simulation
 - Team ranked **40 among 3000+ teams** across the nation participating in eYRC, and among the top 50 teams selected for the **hardware phase**, to implement our algorithms on a Quadcopter

Positions of responsibility

- May 2023 – **Teaching Assistant | Introduction to Quantum Physics (PH107)**
 July 2023 Prof Alok Shukla, Department of Physics
- Conducted tutorial sessions for **40+ students** across various departments
 - Covered content and clarified doubts of the students pertaining to **Quantum Physics**
 - Graded the Midsemester and Endsemester answer sheets for the batch and conducted crib sessions
- May 2023 – **Student Mentor | Mechanical Department Academic Mentorship Programme**
 May 2024 Selected as a part of a 35 membered team out of 125+ applicants based on interviews and peer reviews
- Mentored **5 sophomores**, helping them strike a balance between academics and extracurriculars
 - Acting as the first point of contact in order to actively bridge the gap between faculty and students
- May 2024 – **DCAMP Mentor | Mechanical Department, IIT Bombay**
 Oct 2024 An initiative by the Mechanical department to ensure proper career advice for pre-final year students
- Mentored **4 pre-final year students**, helping them get a better perspective of career opportunities
 - Guided students through their internship season ensuring regular **interactions** and **advice**